



E-commerce Customer Conversion Analysis Report

Identifying Customer Behavioural Patterns, Conversion Drivers and Growth Opportunities

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Report Overview



Understanding Customer demographics, Website engagement behavior, Marketing effectiveness, and the customer purchase journey.

TOOLS

Microsoft Excel: (Operational Analysis)

Power BI Dashboard : (Decision Making Layer)

PostgreSQL (Technical Analysis Layer)



Quick Facts

 **7533**
Total Users

98%
Conversion %

 **7372**
Purchasers

10
Avg pages / user



Executive Summary

The objective of this analysis was to understand reasons for conversion in relation to the high website traffic and marketing activities.

Key Area	Findings
Customer Behaviour	Purchasers show Stronger engagement than Non Purchasers
Marketing	Ads and Discount Increase exposure but do not Gurantee Purchases
Funnel	There isn't Major customer Loss before Final Purchase
Retention	Returning Users represent valuable customers

Business Problems & Objectives



Problem

Although the business achieves strong customer conversion and experiences minimal drop off throughout the purchase Journey, there is a need to understand true drivers behind customer purchases

Identifying what actually influences customer purchases will allow the business to improve profitability by reducing ineffective marketing activities and focusing on high impact customer strategies .

Objectives

- Identify factors influencing purchase Behaviour
- Validate whether marketing activities drive conversion
- Understanding Customer engagement patterns
- Identify Opportunities to improve Profitability

DATA & ANALYTICAL APPROACH

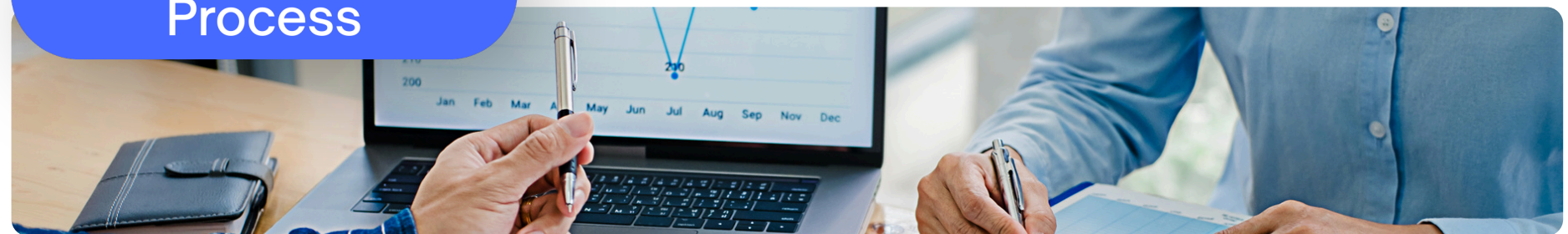


Data Source



- User Demographics
- Website Behavior
- Marketing Interaction
- Purchase Outcomes

Tools & Analytical Process



Excel Analysis

- Pivot Tables
- Charts
- Segmentation
- Trend Analysis

PostgreSQL Analysis

- Aggregations
- Case Statements
- Segmantation Queries
- Funnel Calculations



DATA QUALITY & PREPARATION



Data Cleaning Performed

Using SQL :

- Blank user IDs removed
- Missing Numerical Values treated appropriately
- Missing Categories standardized

Data Formatting

Corrected:

- Numerical Data Types
- Binary Fields

Created

- Purchase Flags
- Customer Segments
- Engagement Indicators

CUSTOMER PROFILE ANALYSIS



Customer Composition

Excel Finding

- Gender Distribution
- Customer type distribution

I.e

- Male customers : 50.09%
- Female Customers: 49.91%

- New Users : 50.05%
- Returning Users : 49.95%

SQL Validation

SQL confirmed

</> SQL

GROUP BY gender

GROUP BY user_type

SQL ANALYSIS



Data Extraction & Validation approach

1 SQL Queries were used to :

- Count total customers
- Identify customer distribution by:
 - Gender
 - Returning vs New users
- Check data consistency and ensure accurate analysis

• Purpose:

To establish a foundation before analyzing customer behaviors .

2 Behavioural Analysis

SQL compared purchasers vs non-purchasers using:

- Average time spent on site
- Average pages viewed
- Bounce rate

Key finding:

- Purchasers demonstrate stronger website engagement compared with non-purchasers.
- User interaction levels provide stronger indicators of purchase intent than basic traffic volume.
- Improving customer engagement can support higher conversion performance.

3 Marketing & Funnel Validation

SQL used:

- CASE statements to create analysis flags
- Aggregations to compare marketing groups
- Funnel calculations to measure customer movement

Analysed:

- Ads clicked vs not clicked
- Discounts seen vs not seen
- User journey:
 - Visitors
 - Engaged users
 - Cart users
 - Purchasers



MARKETING IMPACT ANALYSIS

Marketing Activities increase visibility, but purchase decisions depend heavily on customer engagement and website experience.

Marketing Exposure vs Conversion

Ads	Percentage
Clicked Ads	49.34%
Didn't Click	50.66%
Discounts	
Saw Discount	49.30%
Didn't see Discount	50.70%
INSIGHT:	



CUSTOMER FUNNEL ANALYSIS

TOTAL USERS

 7533



Engaged Users

7499



Cart Users

7435



Purchasers

7372



OVERALL BUSINESS INSIGHTS

- Customer engagement is a stronger purchase indicator than website traffic alone
- Marketing activities generate awareness & require optimization to improve revenue.
- Returning customers provide opportunities for retention strategies .
- Strong funnel performance indicating effective customer journey .
- Opportunities exist that would improve profitability

RECOMMENDATION

1: IMPROVE CUSTOMER ENGAGEMENT :

- Enhance product pages with better descriptions and images.
- Personalise recommendations based on browsing behaviour.
- Improve website navigation to encourage users to explore more pages.

Expected Outcome: Increased customer interaction and sustained conversion performance.

2: OPTIMISE MARKETING SPEND :

- Evaluate campaigns based on conversions and return on investment (ROI)
- Reduce blanket discounts where they do not significantly influence purchases.
- Target campaigns toward high-value customer segments.

3. STRENGTHEN CUSTOMER RETENTION :

- Introduce loyalty or rewards programmes.
- Encourage repeat purchases through targeted offers.
- Send personalised follow-up emails to returning customers.

EXPECTED BUSINESS IMPACT

Recommendation :

- Improve engagement :
- Optimise marketing :
- Improve retention :
- Monitor the funnel :

Expected Benefit :

- Higher customer interaction and sustained conversions
- Reduced unnecessary advertising and discount costs
- Increased repeat purchases and customer loyalty
- Faster identification of performance issues

LIMITATIONS & FUTURE ANALYSIS

LIMITATIONS :

- Analysis was based on the available dataset and variables.
- External influences such as seasonality, competitor activity, or economic conditions were not included.
- The findings reflect historical data and may change as customer behaviour evolves.

FUTURE ANALYSIS

- Analyse customer lifetime value (CLV).
- Investigate product category performance
- Apply predictive models to forecast future purchases.

CONCLUSION

The combined use of Excel and PostgreSQL provided a comprehensive understanding of customer behaviour and purchasing patterns. Excel identified meaningful trends, while SQL validated these findings directly from the raw data, ensuring reliable and evidence-based insights.

The analysis indicates that customer engagement plays a more significant role in purchasing decisions than promotional activities alone. By focusing on improving customer experience, optimising marketing investment, and maintaining an efficient purchase journey, the business can strengthen profitability and support sustainable growth.

THANK YOU

E-commerce Customer Conversion Analysis Report

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Microsoft Excel | PostgreSQL |
Power BI